

Design and Fabrication of Internal Jar Cleaner

1.1 Introduction

Company	Prasiddha Water Industries Pvt. Ltd.
Job title	Graduate Mechanical Engineer
Subject	Design and fabrication of internal jar cleaning machine
Location	Tilottama-9, Rupandehi, Nepal
Project Duration	May-October 2025
Supervisor (Chief Technical Officer)	Roshan Neupane
Degree	Bachelor's Degree in Mechanical Engineering

1.2 Background

1.2.1 Overview

The jar cleaner is designed to efficiently harness rotational force using a powerful DC motor. Its structure is strong and durable, consists of simple and effective design, and delivers efficient cleaning of the jar. All these characteristics make it ideal for the need of the company to cater a machine that makes internal cleaning of the jar easy and effective. SolidWorks is utilized for 3D modeling. It helps in the design of supporting structure, and uniting parts with tight mechanical requirements. ANSYS is applied to conduct structural and modal analysis of the jar cleaner. It examines the structural integrity of the machine along with natural frequency to avoid resonance during operation. This ensures design reliability and structural stability under operational conditions.

The task was to design a machine that can clean the jar internally with efficiency and ease. The ideas regarding the machine design were explored. Cast iron was selected as the material for the structure and standard nuts, bolts and washer were used for assembly. The 3D model was then created in SOLIDWORKS, where the structure was created using L-channel weldments, extrude, mirror, and pattern features. Along with that fasteners were used from the SOLIDWORKS library. The entire model was exported in STEP format. The model was brought to ANSYS for FEA

simulations. Structural analysis and modal analysis were carried out and the deformations and stress concentrations were analyzed.

1.2.2 Objectives

The project motive was to design and fabricate an efficient internal jar cleaner.

Secondary goals were:

- To design strong and durable supporting structure using effective materials.
- To develop a machine that is easy to use and assemble.

1.2.3 Nature of Work

I conducted a thorough analysis of relevant literature and existing market solutions. I selected materials and operating conditions based on engineering needs and performed simulations under operational conditions. I designed the 3D model in SOLIDWORKS and performed structural and modal analysis in ANSYS. I worked with my supervisor to verify the design verification and analyze results, ensuring alignment with company needs. I planned the workflow and set deadlines to ensure consistent progress. I coordinated with multiple vendors for components such as channel rods, fasteners, motor, and sheet metal. I arranged access to workshop space, tools and manpower for fabrication and testing.

1.2.4 Hierarchy

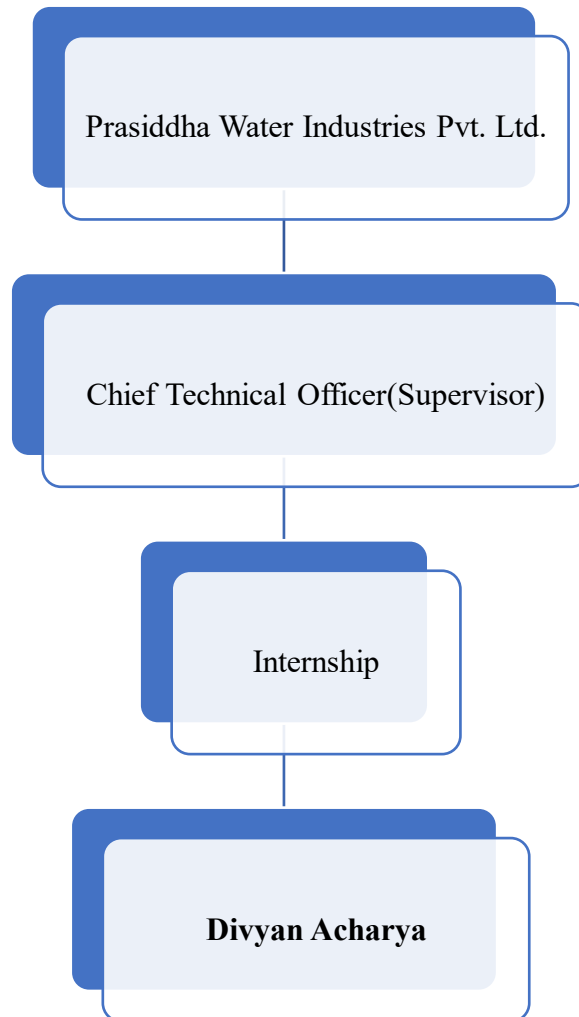


Figure1Hierarchy

1.2.5 Duties

- To research the project requirements for fulfilling the necessary machine required.
- To select appropriate components for high operational efficiency.
- To design the jar cleaner in SOLIDWORKS using weldments, pattern features, extrude and symmetry tools.
- To perform FEA simulations in ANSYS.
- To evaluate result parameters in structural and modal analysis for performance assessment.

1.3. Personal Engineering Activities

1.3.1

I understood the mechanism of problem-solving in the production industry in terms of machine design and testing. I analyzed various kinds of market-relevant machines. I learned how the supporting structure and motor section play a crucial role for overall machine performance and operation. I learned how the specific rotational speed of the motor and geometry of the cleaning tool can increase the efficiency of the internal cleaning process. I acquired ideas related to SolidWorks to make a 3D model of the machine using weldments and other features. I studied the simulation process in ANSYS Mechanical for the study of structural and modal analysis. I studied the mesh refinement and applied accurate boundary conditions for the analysis. I studied various techniques to damp the vibration of the machine. I gained insights on balancing computational accuracy and mesh complexity for improved FEA results.

1.3.2

I nominate cast iron as the structural material due to its high Tensile strength (ultimate) of 700 MPa and excellent vibration damping properties. I took cast iron because its hardness value of around 170–220 HB ensures durability. I chose 370W motor with 4Nm torque, as it provides precise speed with directly coupled connections. I opted to use a rotational speed of 900rpm for efficient cleaning operation. I picked the height of the machine of about 1.1m, length 40cm and breadth of 30cm. I choose L channel (40X40X5mm), pillow block bearing, hollow rod of 30mm, sheet metal with 2.5mm thickness, and standard 12mm nut and bolt. I adopted a simple structural design with a connecting hollow rod for cleaning tool attachment. I chose SOLIDWORKS to model the machine with precision. I used ANSYS for FEA simulations to visualize stress concentration patterns, deformations, and natural frequency of the machine during working conditions.

1.3.3

I began the design of the machine in SOLIDWORKS by creating a new part file and took the top plane for my initial sketch. I used weldments 3D sketch features to draw a sketch for the base structure. I imported a customized L channel for the weldments structural profile feature. I then applied the “Extruded Boss/Base” feature to the components for the assemble of motor and its

connection with the cleaning tool. I chose the “Mirror and pattern” tool to design the cleaning tool. I took a “Fillet” tool to smooth the edges of few designed components. Finally, I saved the complete turbine model in STEP format for further simulation.

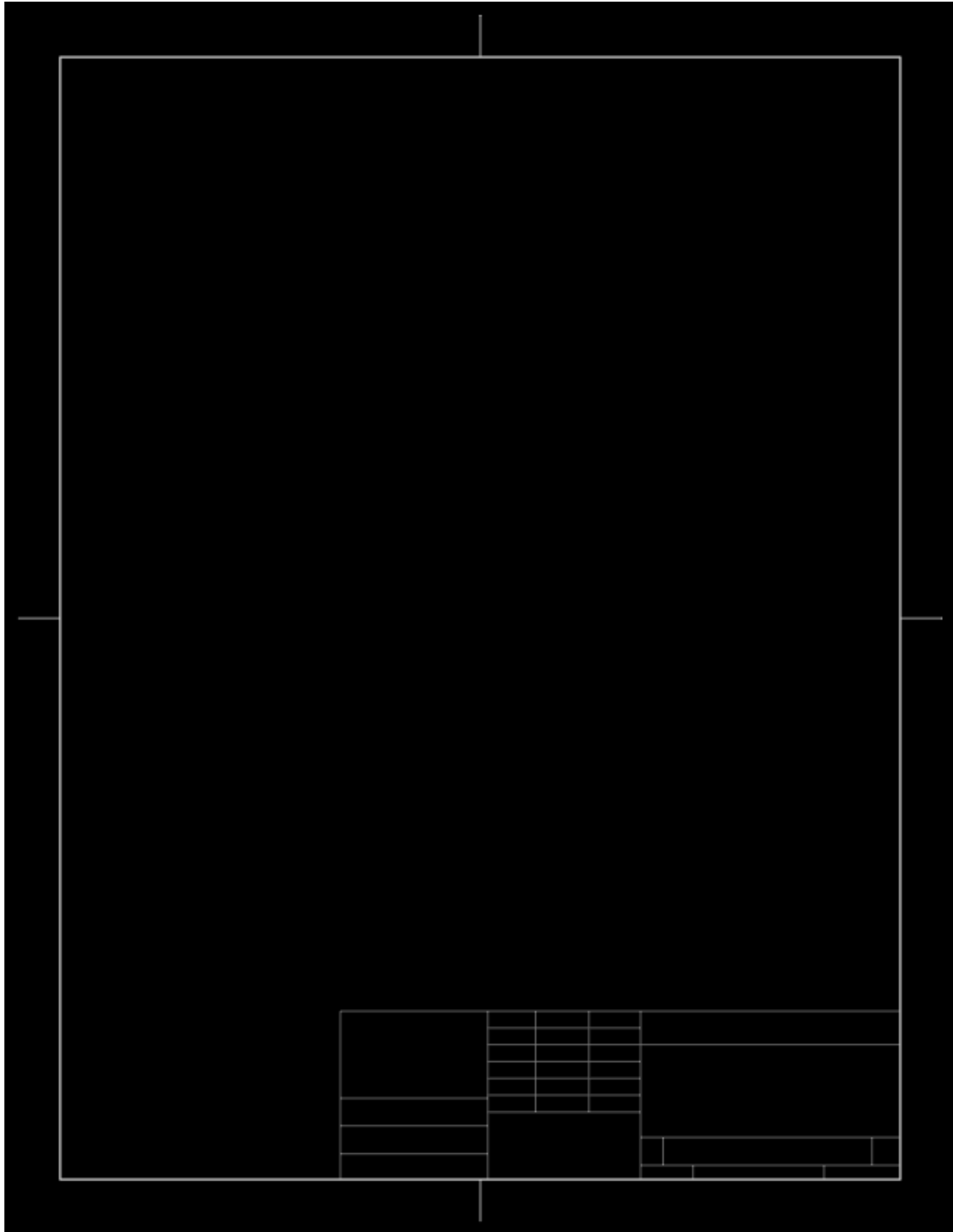


Figure2 Shown a 2D

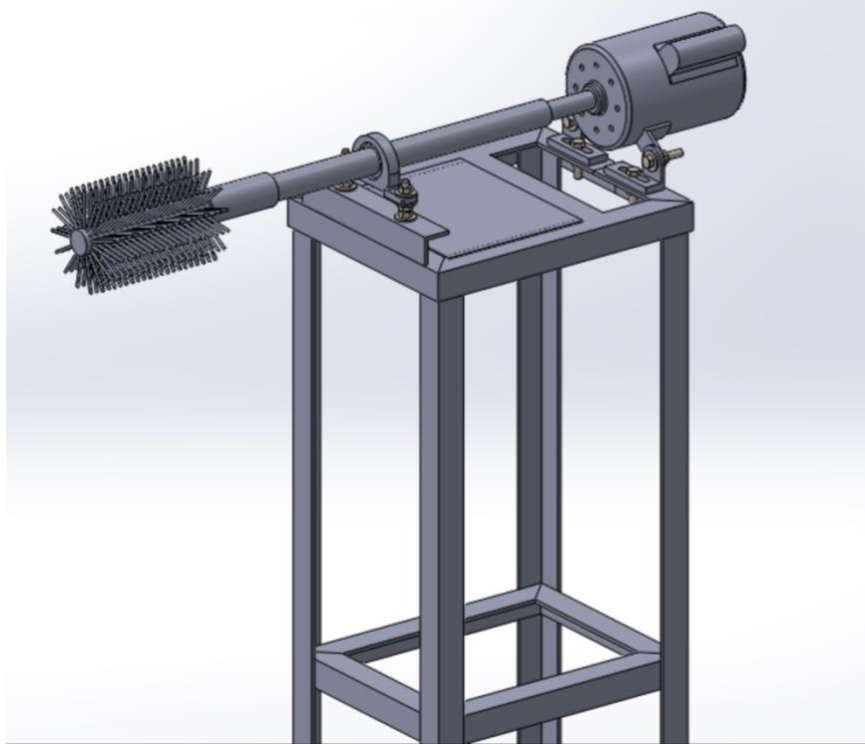


Figure3 Depicted a 3D model

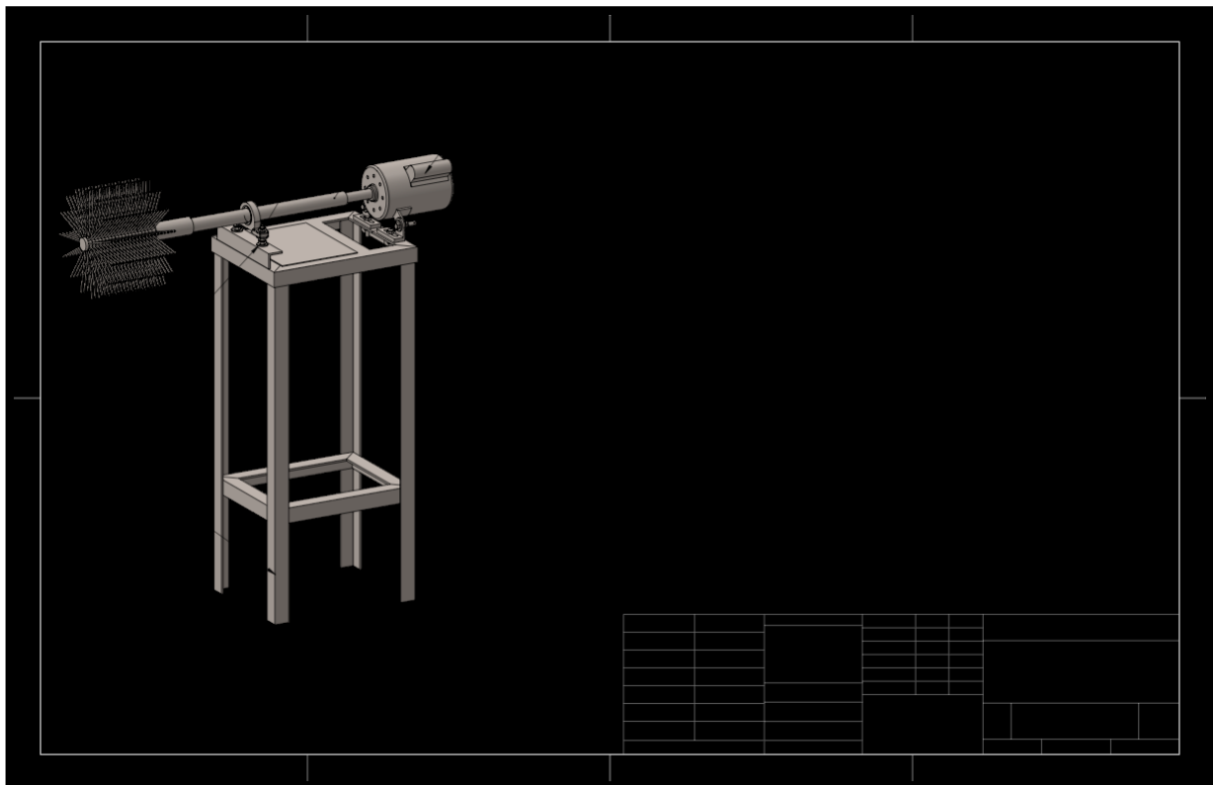


Figure4 Bill of materials

1.3.4

I started the FEA simulation by first importing the STEP model into ANSYS Workbench. I used SPACECLAIM to clean the geometry. I then opened the Meshing module and generated a fine mesh, with 1mm element size using hex-dominant mesh with 35,13,934 nodes and 20,20,426 elements. I assigned the materials to the components. I applied accurate boundary conditions with precisely defining contacts between different parts. I defined necessary results for post processing. I started with static structural analysis and observed the results in-terms of deformation and stress concentrations. I then performed modal analysis to visualize different 6 natural frequency along with their deformations and stress concentrations. I refined the mesh after multiple iterations of analysis for more accurate results. I analyzed all the necessary parameters for complete and accurate analysis of the machine.

1.3.5

I purchased all the necessary equipment for the fabrication and collected it in the fabrication workshop. I started with the supporting structure by precisely cutting the L channel using angle grinder with metal cutting disc. I choose to weld the structure using Shielded Metal Arc Welding (SMAW). I used the similar technique for other components as well. I choose a connector hollow rod of 400mm with interference fit with the motor shaft. I used the 14mm drill bit to drill holes in areas where nut and bolt is used for the connection.

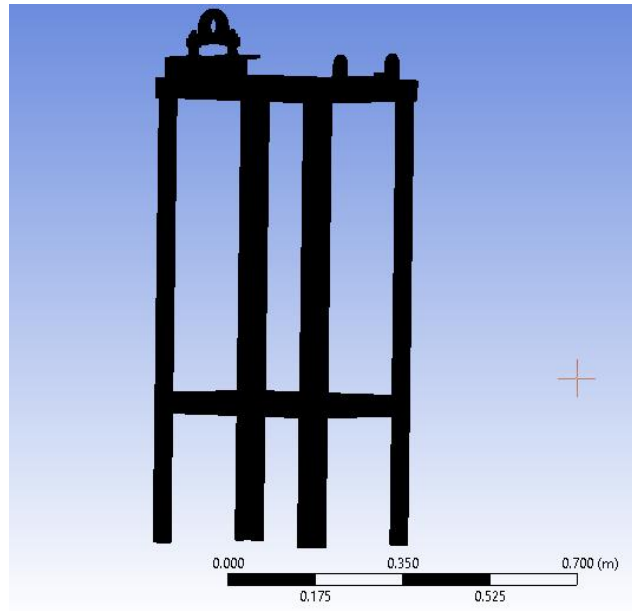


Figure5 Meshed view of the model

1.3.5

I started with a static vertical bending analysis with a fixed support at the base and one of the legs connected to the supporting structure at the factory. Along with this, an applied force of 100 N in the direction of acceleration due to gravity was considered. The force represents the mass of the motor. I found the maximum equivalent stress to be 5.57 MPa, and the maximum total displacement recorded was 0.2 mm. The modal analysis recorded six rigid-body modes of vibration at 0 Hz, 26.882 Hz, 35.146 Hz, 68.081 Hz, 93.769 Hz, and 122.47 Hz. The operating frequency of the motor was 15 Hz. I concluded that the structure is sufficiently rigid and damped for operation with maximum efficiency under the given operational conditions.

Result

For Equivalent (Von-mises) Stress

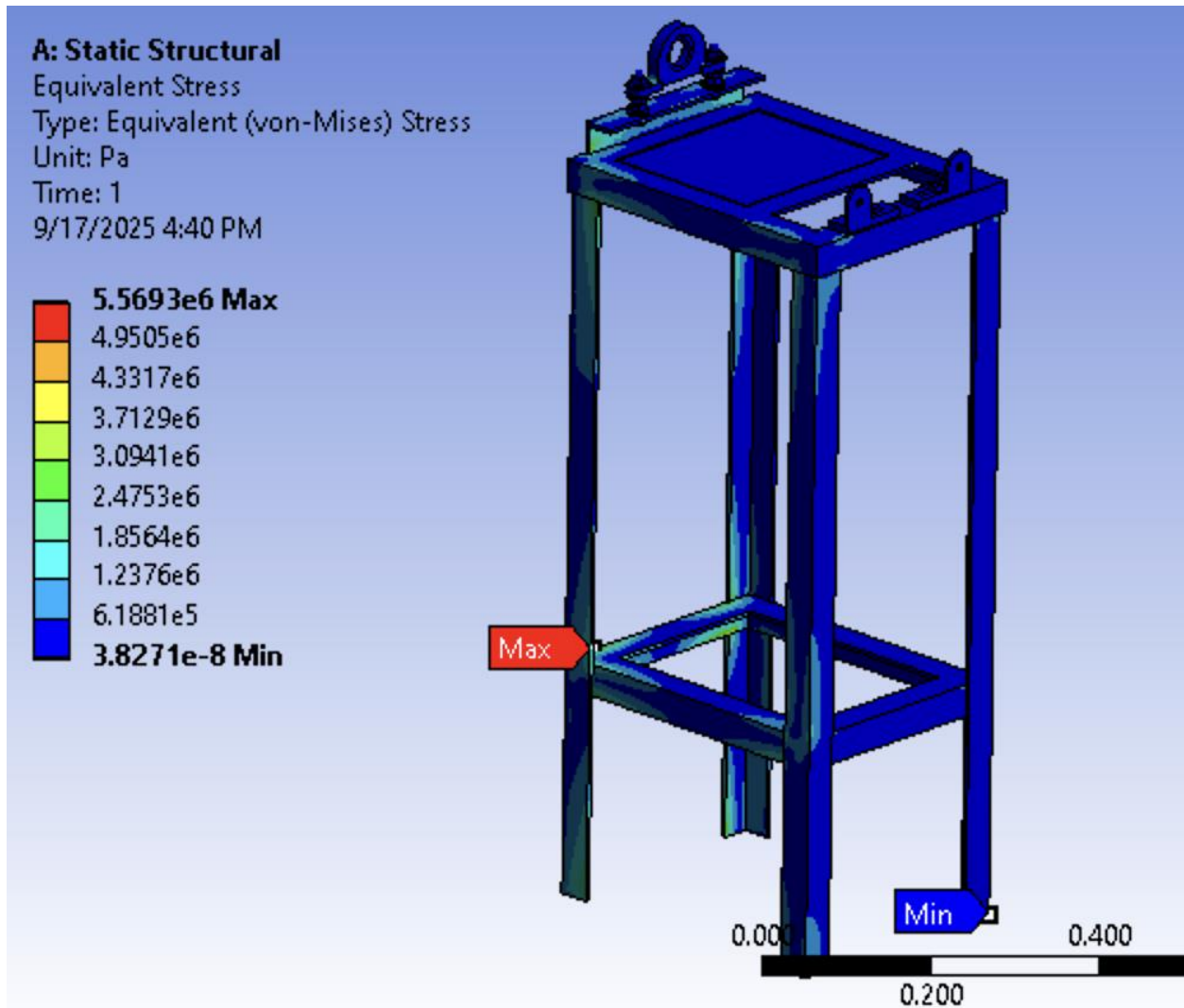


Figure6 Equivalent Von-mises stress under given boundary conditions

Maximum Equivalent Stress = 5.57MPa

Minimum Equivalent Stress = 0 Mpa

Since the maximum equivalent stress doesnot exceed the ultimate tensile strength, the structure will not lead to failure at any point of operation.

For Total Deformation

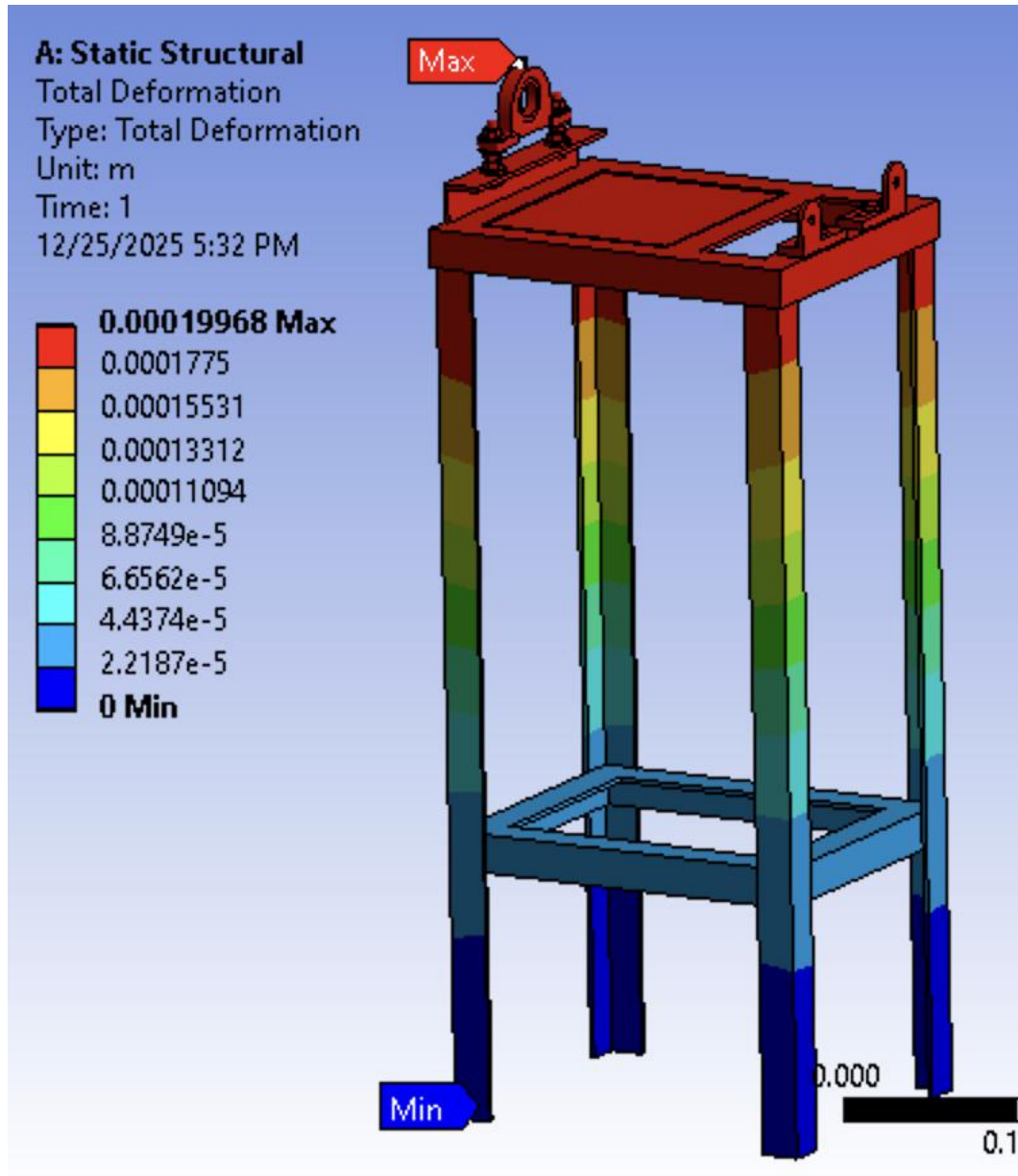


Figure7 Total Deformation of the system under given boundary conditions

Maximum Total Deformation = 0.2mm

Minimum Total Deformation = 0mm

The maximum deformation is comparatively very low within the elastic limits.

Modal Analysis

	Mode	☒ Frequency [Hz]
1	1.	0.
2	2.	26.882
3	3.	35.146
4	4.	68.081
5	5.	93.769
6	6.	122.47

Figure8 Six Rigid-body modes of vibration

All the above modes of vibrations do not interact with the natural frequency of the motor, thereby preventing resonance in the system.

1.4 Problem and Solution

I faced an issue during motor selection and design configuration, because the machine needed to be easy to operate and the rotational speed along with the cleaning tool attachment need to operate in smooth operation for effective cleaning. I made sure that the speed of the selected motor is enough for the efficient operation without any need of gearbox or belt pulley mechanism. I realized that during operation vibration might cause huge problem during the working of the motor while cleaning. I found that if the base of the structure is attached to the ground along with vertical channel in the structure is welded with the column of the truss in the factory itself. Altogether, these things will damp most of the vibrations of the machine. I went to my supervisor and explained the overall selected and designed components for the machines. I was asked to go back to selecting modify and try with other attachment points for more stable operation of the machine by reducing vibrations. I realized the adding extra attachment points along helps to improve damping in the machine.

1.5 Creative Works

I tested the jar with specific different amount of water along with the different variable speed motor while selecting the motor for constant speed motor without any speed reduction mechanism. I designed a horizontal adjustable design for the motor placement and vertical height adjustable design for the pillow block bearing placement to couple the overall working component of the machine. I solved the vibrational issue of the machine by providing base attachment along with a side attachment. I enhanced the result accuracy by precisely assigning boundary conditions during the FEA simulations.

1.8 summary

1.8.1

The project was attained by successfully designing and fabricating an internal jar cleaner. The mechanism and operation of machine were investigated in the beginning. SolidWorks was employed to create the 3D model of the assembly with the incorporation of features weldments, mirror, pattern, and Extruded Boss/Base and cut techniques. The developed model was brought

into ANSYS, where material assignment, mesh generation, connection setup, and FEA boundary conditions were established. A bolted connection with the floor along with a welded connection to the truss support column was made for damping the vibrations during the rotational moment of the cleaning tool. The machine functioned best when it was working with good amount of water inside the jar and the placement of the jar is concentric to the cleaning tool. It achieved the aims through proper design modeling, analysis and testing.

1.8.2

I developed strong modeling skills in SOLIDWORKS and learned to simulate structural and modal analysis in ANSYS. I improved my FEA skills and gained hands-on experience with simulation and actual testing results analysis for effective design. I manage to transform the 3D model into a functional product, incorporating all the necessary design features and operational capabilities through effective problem solving. I also enhanced my knowledge on techniques to reduce vibration and its effects in any machinery. I learned about problem-solving, coordination and technical communication skills.